

Jie Kang

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Research Interests:

Large Language Models, Parameter Generation, and Graph Machine Learning.

I am interested in understanding how neural networks adapt under dynamic environments, with a focus on reasoning, memory, and parameter-efficient adaptation.

Education

University of Virginia, Charlottesville, VA

Aug 2025 – Present

M.E. in System Engineering, GPA 4.0 / 4.0

Coursework: ML on Graphs, NLP, Probabilistic Machine Learning, Statistical Modeling, Optimization

East China Normal University, Shanghai, China

Sep 2018 – Jun 2021

M.S. in Communication and Information System, GPA 3.1 / 4.0

Research Focus: Graph Neural Networks, Complex network dynamics, Deep learning

Nanjing Forestry University, Nanjing, China

Sep 2014 – Jun 2018

B.S. in Electronic Information Engineering, GPA 3.6 / 4.5

Research Experience

Graph-based Memory Evaluation for Multi-Agent Systems

Jan 2026 – May 2026

ML on Graphs Course Project, University of Virginia

- Proposed SeqMem-Eval, a systematic evaluation protocol for sequential LLM memory systems, introducing five metrics: online utility, hold-out generalization, backward transfer, forgetting, and efficiency.
- Benchmarked G-Memory against 7 baselines across 6 datasets using Qwen3-8B and MiniMax-M2.7.
- Identified graph-structured memory does not yield consistent performance improvement under sequential inference, revealing a gap between memory utilization and memory evolution.
- Analyzed computational bottlenecks, showing BFS-based retrieval dominates efficiency cost, motivating scalable memory design.

Deep Learning for Epidemic Threshold Prediction on Complex Network

Sep 2018 – Jun 2020

Graduate Researcher, East China Normal University

- Developed learning-based frameworks to model dynamics on complex networks, focusing on identifying critical transition points in nonlinear systems (e.g., epidemic thresholds).
- Proposed a hybrid supervised/unsupervised learning approach ("confusion scheme") to detect phase transitions, connecting learning dynamics with underlying system behaviors.
- Applied geometric representation learning (Poincaré embedding) to capture hierarchical structures and effectively mitigate power-law degree imbalance in scale-free networks.

- Achieved 95%+ accuracy in identifying thresholds of disease spread(SIS/SIR dynamics) across both synthetic and real-world empirical networks.
- Presented an oral report at the 15th National Complex Network Conference (2019).

COVID-19 Spreading Dynamics Modeling

Dec 2019 – Jun 2020

Research Collaborator, East China Normal University

- Developed a non-Markovian dynamic system model with latent states to capture complex temporal evolution in real-world data.
- Designed parameter inference framework via optimization (Levenberg–Marquardt), enabling robust estimation under incomplete observations.
- Concluded that non-negligible hidden population could persist post-pandemic, potentially leading to long-term community disease.
- Co-authored two papers (*arXiv* and *medRxiv*, 2020)

Professional Experience

Noah Holding Co. Ltd., Shanghai, China

Jun 2021 – Aug 2024

Big Data Center – Data Analyst

- Developed machine learning models for targeted marketing system, serving 500K+ users.
- Applied clustering algorithms to segment customers, increasing conversion rates by 20%.
- Deeply involved in digital transformation initiatives impacting 2000+ employees.
- Designed automated data pipelines and real-time dashboards for monitoring key metrics across multiple products.

Microsoft, Shanghai, China

Dec 2019 – Sep 2020

Dynamics 365 - Support Engineer Intern

- Analyzed performance data for 50+ engineers by calculating KPIs like cases closed per week, satisfaction scores.
- Specialized in Dynamics 365 Finance & Operations application in ERP domain.
- Contributed to a Hackathon Project, developing an image recognition system for a Self-managed Cabinet.

Publications

[1] Qi Ni, **Jie Kang**, Ming Tang, Ying Liu, Yong Zou. “Learning epidemic threshold in complex networks by convolutional neural network.” *Chaos: An Interdisciplinary Journal of Nonlinear Science*, **29**, 113106 (2019). DOI: 10.1063/1.5121401

[2] **Jie Kang**, Ming Tang. “Supervised Learning Epidemic Threshold of SIR Model in Complex Networks.” *Proceedings of the 4th International Conference on Wireless Communications and Applications (ICWCA)*, 2020.

[3] Yong-Shang Long, Zheng-Meng Zhai, Li-Lei Han, **Jie Kang**, et al. “Quantitative assessment of the role of undocumented infection in the 2019 novel coronavirus (COVID-19) pandemic.” *arXiv:2003.12028* (2020).

[4] Zheng-Meng Zhai, Yong-Shang Long, **Jie Kang**, et al. “State-by-State prediction of likely COVID-19

scenarios in the United States and assessment of the role of testing and control measures.” *medRxiv* (2020).

Skills

Programming: Python (PyTorch, NetworkX, HuggingFace Transformers, scikit-learn), R, SQL, C/C++, LaTeX

ML / AI: Graph Neural Networks, CNN, LLM evaluation & benchmarking, multi-agent systems, graph representation learning, time-series modeling

Tools: Tableau, Git

Awards & Honors

- **2nd Prize**, ‘HUAWEI Cup’ 16th China Post-Graduate Mathematical Contest in Modeling (2019)
- **Outstanding Graduate**, Nanjing Forestry University (2018)
- Merit Student & Outstanding Student Cadre Scholarships, Nanjing Forestry University (2014-2018).
- Academic Scholarships, East China Normal University (2019-2021)
- Host, Jiangsu Students’ Innovation & Entrepreneurship Training Program — **Final Excellent Acceptance** (2015–2017)